

Verb-vs-sink coverage table

Step 1 of the foundation slab from [north-star.md](#) §8. Living artefact — when verbs or sinks change, update this file.

The principle (from [north-star.md](#) §4): **consistent shape wins over consistent implementation**. Cells use the same vocabulary regardless of sink; each sink consumes what it understands and ignores the rest. Holes are honest and fall out for free.

Legend

Symbol	Meaning
D	Direct — the sink natively understands the verb (built-in wire mapping)
M	Mappable — sink consumes via per-rig config (CC table, CV jack assign)
—	No-op — silently dropped (no meaningful interpretation on this sink)
G	Global — affects the engine, not a single voice
?	Unknown / not yet decided

The **M** cells are where the per-rig translation layer ([north-star.md](#) §8 step 6) lives. They're configuration data on the Studio side, not code on the sink side.

The — cells are the price of consistent syntax. A cell that writes `# vowel "a"` against a MIDI drum kit silently ignores the vowel — and that's correct.

Sinks under consideration

Column	Wire format / process
SuperDirt	<code>/dirt/play</code> OSC bundles → localhost:57120 (sample player)
MIDI generic	Note-On/Off + CCs on a CoreMIDI destination, single channel
MIDI drumkit	Note-On per slot (sound → MIDI-note map), single channel
FH-2 daemon	SysEx writes through <code>~/fh2/control.sock</code> (polysignal config)
CvRouter	OSC to cv-router → ES-9 V/oct + Gate jacks
Yarns poly	MIDI multi-channel poly mode (future)

Coverage table

Verb (cell-side)	SD	MIDI generic	MIDI drumkit	FH- 2	CvRouter	Yarns poly
Source						
note	D	D	—	—	D	D
n	D	—	—	—	—	—
sound	D	—	D (slot map)	—	—	—
Amplitude						
gain	D	D (→ velocity)	D (→ vel)	—	M (→ vel-CV)	D (→ vel)
velocity	D	D	D	—	M	D
pan	D	M (CC10)	M (CC10)	—	M	M (CC10)
speed	D	—	—	—	—	—
accelerate	D	—	—	—	—	—
Envelope						
attack	D	M (CC73)	—	—	M (env-CV)	M (CC73)
release	D	M (CC72)	—	—	M (env-CV)	M (CC72)
legato	D	D (note-len)	—	—	D (gate-len)	D (gate-len)
sustain	D	M (CC?)	—	—	—	M
Filters						
lpf	D	M (CC74)	—	—	M (aux-CV)	M (CC74)
lpq	D	M (CC71)	—	—	M	M (CC71)
hpf	D	M	—	—	M	M
hpq	D	M	—	—	M	M
bpf	D	—	—	—	—	—
bpq	D	—	—	—	—	—
Effects						
room	D	M (CC91)	—	—	—	M (CC91)
size	D	M (CC?)	—	—	—	M
delay	D	M (CC?)	—	—	—	M
delaytime	D	—	—	—	—	—
delayfeedback	D	—	—	—	—	—

Verb (cell-side)	SD	MIDI generic	MIDI drumkit	FH-2	CvRouter	Yarns poly
crush	D	—	—	—	—	—
shape	D	—	—	—	—	—
vowel	D	—	—	—	—	—
Pitch / timing						
coarse	D	M (PB)	—	—	D (semitone-CV)	M (pgm)
cut	D	M (note-off)	D (cut group)	—	D (gate-off)	D
nudge	D	D (schedule δ)	D (schedule δ)	D	D	D
begin	D	—	—	—	—	—
end	D	—	—	—	—	—
loop	D	—	—	—	—	—
Global						
cps / bpm	G	G	G	G	G	G
unit_	D	—	—	—	—	—
orbit	D	—	—	—	—	—
Sufflamen only						
polysignal record	—	—	—	D	—	—
Vetula only						
chord / voicing	(via N)	(via N)	—	—	(via N)	D

"via N" = via the **Notation**-shaped expansion of a chord into N parallel note events; each note event then follows the row's column rules.

How to read this table

- A row tells you "which sinks understand this verb."** **gain** is D-or-M everywhere except FH-2 (which is a config sink, not a note emitter). **vowel** is D on SD only — that's a SuperDirt-specific formant filter. Cells writing **# vowel "a"** against any other sink silently no-op.
- A column tells you "what this sink accepts."** SD's column is almost all D — it's the canonical sample-player target and the vocab was designed against it. CvRouter's column is mostly M —

modular synthesis requires per-rig patching decisions. FH-2's column is almost all — — it doesn't emit events at all, it consumes polysignal config records.

- **The M cells are work, but configurable.** Each M cell becomes one line in a Studio-level destination declaration:

```
ableton_piano1 = midiInstrument
{ channel: 5
  , ccMap:
    { lpf:      CC 74
      , room:   CC 91
      , attack: CC 73
      , release: CC 72
      , pan:    CC 10
    }
  , defaults: { velocity: 90 }
}
```

The cell that writes `# lpf 800 # room 0.4 # gain 0.7` then emits the right CC frames at runtime, no sink-side code change required.

- **The — cells are the price of consistent syntax.** We deliberately don't make `vowel` work on a MIDI drum kit by some contrived mapping. If you want vowel-filter formant modulation, route to a sink that has it (SD) or to a soft-synth whose formant CC you've mapped via the **M** layer.

What this table is missing (TODO)

- **Sufflamen-specific verbs.** Once Sufflamen's typed PolySignal record is firmed up (currently in `Tidal.Polysignal*` modules under task #59), add its verb surface as a separate row group.
- **Vetula-specific verbs.** Once Vetula lands (task #134), add its `voicing`, `progression`, `leading` etc. as a row group, with the caveat that they expand to N parallel `note` events per chord.
- **Live-control bus verbs.** `live "name"`-style reads are cross-cutting and not per-sink; document separately when the typed-bus work (task #150) lands.

Provenance

Initial population from a survey of `src/Tidal/Controls.purs` (atlantis, 14 verbs) + the wider SuperDirt vocab implemented in `purperl-tidal-port` (34 verbs, the lpf/hpf/room/vowel family expansion), 2026-05-22. Sinks surveyed from `src/Tidal/Dispatch/`, `src/grids_voice.erl`, `src/rene_voice.erl`, `src/repetitor_voice.erl`, and `src/virtual_polysignal_voice.erl`.

The wider SD vocab (lpf/hpf/room/attack/etc.) is in `purperl-tidal-port` but not yet in atlantis. Either port back, or add new SD-only verbs incrementally as cells call for them.